Moreton Bay Seagrass Resilience Modelling

DRFA 21/22 Biodiversity Conservation Marine Final Report Project

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2026-05-11

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1 Introduction

Over the past few decades, various scientific studies have informed our current understanding of trends, behaviors, and flood responses of seagrass cover in Moreton Bay [Abal and Dennison, 1996, Udy and Dennison, 1997, Samper-Villarreal et al., 2016, Ovsyanikova et al., 2026] as well as Australia more widely [Kilminster et al., 2015, Gilby et al., 2018, O'Brien et al., 2018]. Remote sensing has offered new capabilities for seagrass cover mapping [Roelfsema and Phinn, 2009], while innovative statistical approaches have offered novel methods for studying resilience at site-specific scales [Gilby et al., 2023]. Our work adds to this body of literature by applying spatially explicit statistical models [Cressie, 1993] to study the drivers, resilience, and recovery of total percent seagrass cover in Moreton Bay and Wide Bay [Dumelle and Peterson, 2026]. Specifically, we provide and contextualize answers to three specific research questions, applicable to Moreton Bay:

1. Can we identify water quality covariates that act as triggers for seagrass cover management responses? Are these triggers different based on whether they result from a specific event (e.g., flood) or ambient conditions?
2. Did seagrass cover improve after the flood event(s)? If so, how and where?
3. Can we identify site-specific resilience indicators, which describe where recovery is likely to be enhanced or impeded? If so, what factors contribute to seagrass resilience?

2 Seagrass Cover Model

2.1 Methods

2.1.1 Data

The seagrass cover dataset was provided by Science Under Sail Australia (SUSA; https://www.scienceundersail.com.au/). It included 7657 observations across three financial years (2020-2021, 2022-2023, and 2023-2024), where water quality and geographic covariates were also available. Of the 7657 observations, 1995 were collected in 2020-2021 (pre-flood), 3329 in 2022-2023 (1-year post-flood), and 2333 in 2023-2024 (2-years post-flood) (Table 1). There were six seagrass species found in the dataset: *Halophila decipiens* (HD), *Halophila spinulosa* (HS), *Halophila ovalis* (HO), *Halodule uninervis / Zostera muelleri* (ZMHU), *Syringodium isoetifolium* (SI) and *Cymodocea serrulata* (CS).

Seagrass cover is a percentage ranging from 0 (no seagrass) to 100 (complete seagrass cover). Total cover tends to be highest near the eastern shoreline of the Bay and lowest in the middle (Figure 1). Note that 2022-2023 (post-flood) had the the largest number of seagrass cover samples, as well as the only samples collected in the middle of the Bay.

Table 1: The sample size and percent of total sample size for total percent seagrass cover observations in Moreton Bay, by financial year.

Financial Year	Sample Size	Percent of Total
2020-2021	1995.00	26.1%
2022-2023	3329.00	43.5%
2023-2024	2333.00	30.5%
Total	7657.00	100.0%

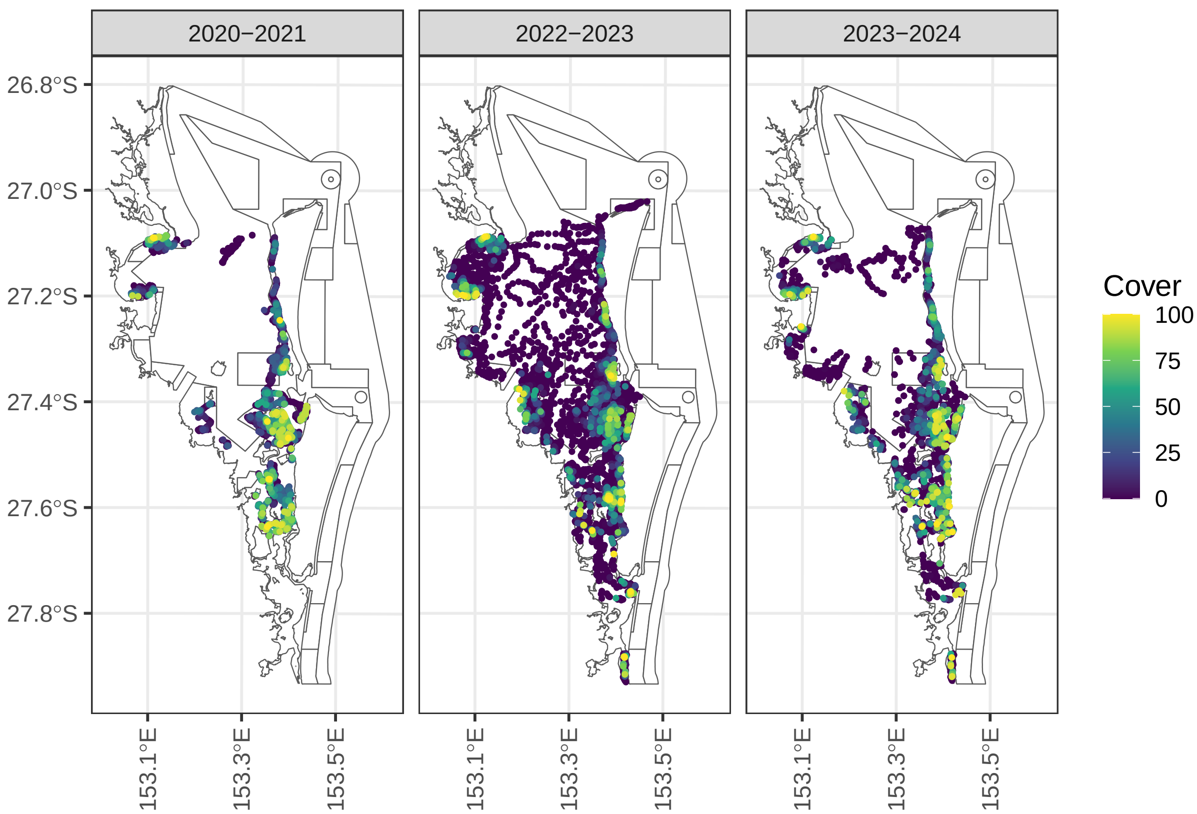

Figure 1: Total percent seagrass cover in Moreton Bay by financial year.

EcoFutures Consulting used eReefs data [Steven et al., 2019] to generate a broad suite of ambient and flood-related turbidity, salinity, and temperature variables at biologically and ecologically relevant temporal windows (Table 2). Ambient water quality variables were based on the mean, minimum, and maximum values for three periods: 7, 30, and 90 days prior to sampling. Ambient variables were also calculated representing the frequency of days water quality thresholds were exceeded (above or below) and the maximum spell (i.e., duration) of days those thresholds were continuously exceeded during the time periods. Similar water quality variables representing flood-event conditions were also calculated for a 30-day flood period.

Additional covariates were provided by EcoFutures Consulting representing geographic features, proximity to rivers, and depth, as well as the years since the flood and seagrass growing season. These variables have the following structure:

- Sediment Type includes six categorical levels: Sand, Sandy Mud, Muddy Sand, Mud, Rock, and Other.
- Tidal Range has three categorical levels: Shallow Subtidal, Intertidal, and Deep Subtidal.
- Years Since Flood is numeric ranging from zero (pre flood) to 1.93 years post-flood (February, 2024).
- Nearest River includes six categorical levels: Brisbane River, Caboolture River, Logan River, Ningi Creek, Pimpama Coomera and Nerang Rivers, and Pine River.
- Distance to (Nearest River) Outlet is numeric and ranges from 0.24 to 34.97 kilometers.
- Growing Season is a categorical variable with two levels: Growing (September - December) and Not Growing (January - August).
- EHMP subzone has eleven categorical levels: Bramble Bay, Broadwater, Central Bay, Deception Bay, Eastern Banks, Moreton Bay Open Coastal Waters, North-Eastern Moreton Bay, Northern Banks Open Coastal Waters, South-Eastern Moreton Bay, Southern Bay, and Waterloo Bay (Figure 2).

2.1.2 Exploratory Analyses

Extensive exploratory analyses were undertaken to 1) understand the data structure, 2) visualize spatial and temporal patterns in the the data, 3) identify outliers/errors, 4) examine correlations between potential covariates and the seagrass cover, as well as multicollinearity among covariates, 5) compare the fit of various models, and 6) evaluate model assumptions.

Results revealed substantial collinearity among ambient and flood-related water-quality variables, as expected given their nested temporal windows and threshold values. We retained only one variable from each correlated group and prioritized sediment type, tidal range, turbidity, temperature, and time since flood when making these decisions. We also removed other variables showing no meaningful association with seagrass cover or where the relationship lacked biological plausibility. This included all of the salinity variables, which had little variability and did not exhibit strong relationships with seagrass cover. Geographic variables exhibited

Table 2: Moreton Bay water quality variables considered in the exploratory analyses, including the variable name and measurement unit (in parentheses), the type of water-quality variable (Ambient, Flood), the time period in days, statistic calculated, and the threshold value used (if any).

Variable	Type	Period	Statistic	Threshold
Turbidity (NTU)	Ambient	30, 7, 90	Mean	
Turbidity (NTU)	Ambient	30, 7, 90	Frequency	> 3
Turbidity (NTU)	Ambient	30, 7, 90	Frequency	> 5
Turbidity (NTU)	Ambient	30, 7, 90	Frequency	> 6
Salinity (PSU)	Ambient	30, 7, 90	Frequency	< 20
Salinity (PSU)	Ambient	30, 7, 90	Frequency	< 25
Salinity (PSU)	Ambient	30, 7, 90	Frequency	< 30
Temperature (deg C)	Ambient	30, 7, 90	Mean	
Turbidity (NTU)	Flood	30	Mean	
Turbidity (NTU)	Flood	30	Max	
Turbidity (NTU)	Flood	30	Frequency	> 3
Turbidity (NTU)	Flood	30	Frequency	> 5
Turbidity (NTU)	Flood	30	Frequency	> 6
Turbidity (NTU)	Flood	30	Duration	> 3
Turbidity (NTU)	Flood	30	Duration	> 5
Turbidity (NTU)	Flood	30	Duration	> 6
Salinity (PSU)	Flood	30	Min	
Salinity (PSU)	Flood	30	Mean	
Salinity (PSU)	Flood	30	Frequency	< 20
Salinity (PSU)	Flood	30	Frequency	< 25
Salinity (PSU)	Flood	30	Frequency	< 30
Salinity (PSU)	Flood	30	Duration	< 20
Salinity (PSU)	Flood	30	Duration	< 25
Salinity (PSU)	Flood	30	Duration	< 30

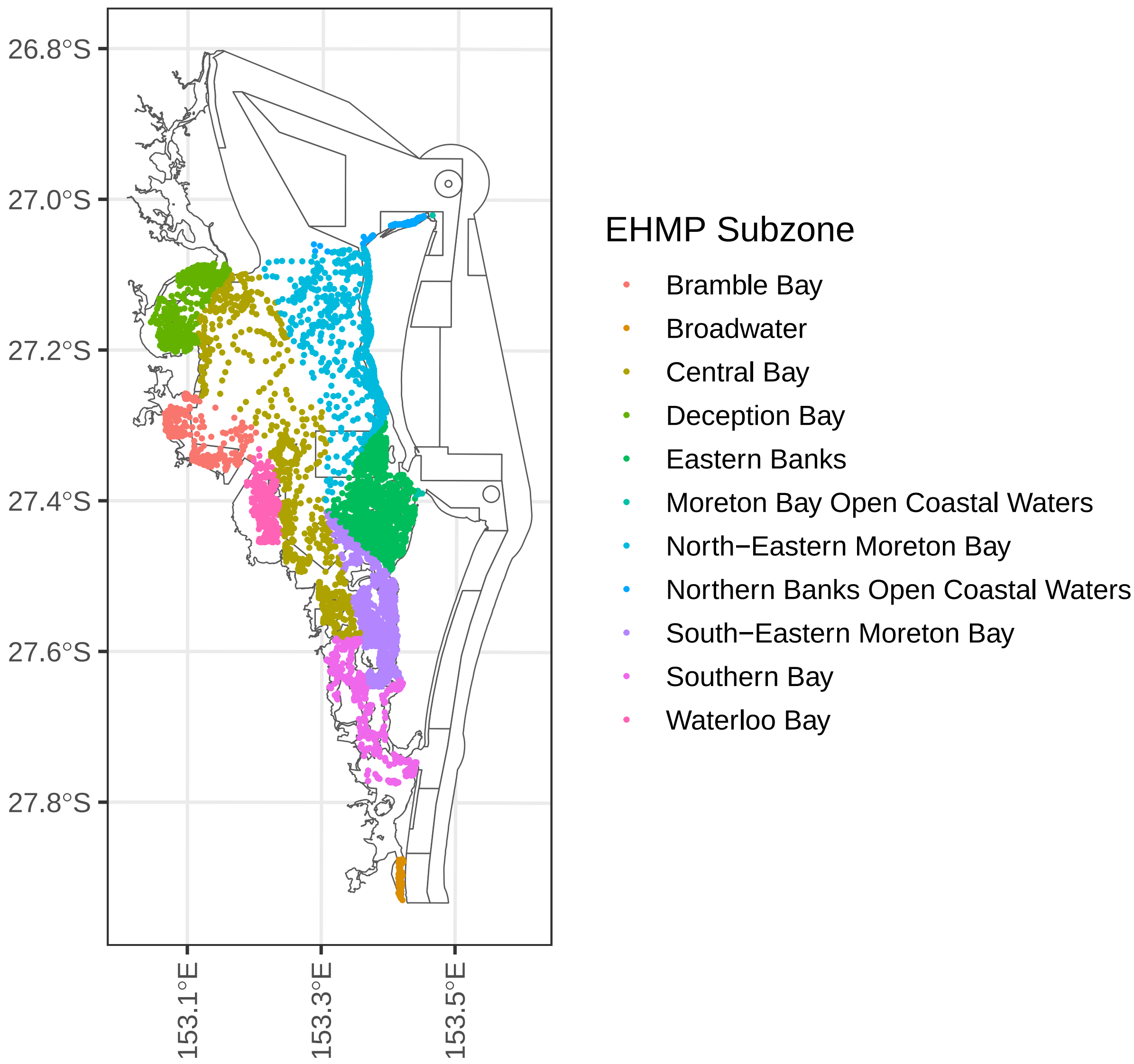

Figure 2: Observations and their EHMP Subzones in Moreton Bay.

less collinearity than water quality variables; however, we removed the distance to the nearest river outlet and the nearest river variables due to weak associations with seagrass cover after accounting for other model covariates. Growing season was excluded because a simple monthly proxy inadequately captures seasonal seagrass growth dynamics.

The final set of potential covariates considered in the modelling included mean turbidity in the 90 days prior to sampling, mean turbidity during the flood event, mean temperature in the 90 days prior to sampling, sediment type, tidal range, years since flood, and EHMP subzone.

2.1.3 Statistical Modeling

We considered multiple modeling approaches in this study, which each accounted for the data distribution, nonlinear relationships between seagrass cover and covariates, and spatial dependence inherent in biological data collected through space and time to different degrees (Appendix 1). We used the full set of potential covariates to fit a nonspatial linear model, spatial linear model (i.e., spatial model), spatial beta regression model, spatial gamma regression model, and a random forest model, comparing the model fit of each.

There were several reasons for fitting these initial models. The nonspatial linear model provided a useful baseline to compare the spatial linear model against, elucidating the impact of spatial dependence on model fit. The spatial beta regression model is a generalized linear model used to characterize bounded proportional data ranging between zero and one. To adhere to this restriction, total seagrass cover was 1) scaled to a proportion and 2) if exactly zero or one, set to 0.01 or 0.99, respectively. The spatial gamma regression model is a generalized linear model used to characterize skewed positive data. To adhere to this restriction, a value of 0.01 was added to seagrass cover observations that were equal to zero. Finally, the random forest model provided a useful assessment of a machine learning approach, which requires fewer distributional assumptions than the aforementioned statistical approaches and more naturally describes nonlinear relationships between variables (Appendix 1). We also characterized random forest effects using partial dependence plots [Greenwell, 2017], which isolate the marginal contribution of a variable while holding other variables at their observed values. Out-of-sample predictions were generated for the four statistical models using leave-one-out cxross validation, while out-of-bag validation was used for the random forest model. These out-of-sample predictions were then used to calculate the mean bias (MBias) and root-mean-square-prediction error (RMSPE) for each model.

The results of this initial model comparison showed that the spatial linear model had the lowest RMSPE, followed by the random forest, generalized linear models, and the nonspatial linear model (Table 3). Mean bias (MBias) tended to be small relative to root-mean-squared-prediction error (RMSPE). The nonspatial and spatial linear models also had much lower mean bias values than the random forest model, while the two generalized linear models produced unacceptably biased predictions. Nevertheless, all five approaches identified similar relationships between seagrass cover and the covariates, in the directions expected. Random

Table 3: Moreton Bay out of sample prediction metrics for the five modeling approaches, including mean bias (MBias) and root mean square prediction error (RMSPE).

Approach	MBias	RMSPE
Spatial Linear Model	-0.02	18.41
Random Forest	-0.11	18.84
Spatial GLM - Beta	-2.76	19.60
Spatial GLM - Gamma	5.42	20.07
Linear Model	0.00	22.54

forest partial dependence plots [Greenwell, 2017] suggested that linear effects of the model variables were generally reasonable. We chose to use a spatial linear modeling framework for the Moreton Bay seagrass model based on these results and the ease of interpretability.

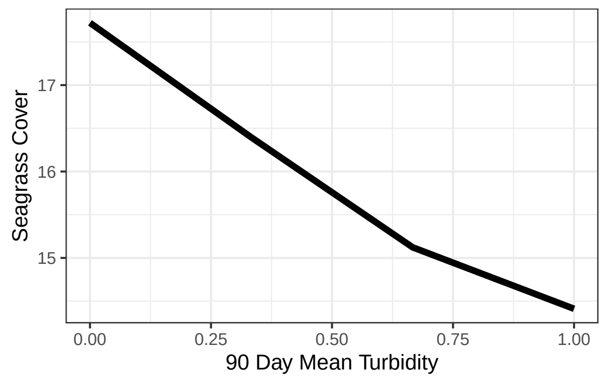

Figure 3: Random forest partial dependence for seagrass cover as a function of mean turbidity in the 90 days prior to sampling.

A suite of spatial linear models (Appendix 1, Equation 1), with exponential spatial covariance functions and a random effect for EHMP subzone, were fit using the `splm()` function in `spmodel` Dumelle et al. [2023]. Parameters were estimated using restricted maximum likelihood and models primarily compared using leave-one-out cross validation; though AIC or BIC could also be used if parameters were estimated using maximum likelihood. In addition, the relationships between seagrass cover and covariates needed to be plausible.

The final total seagrass cover model for Moreton Bay included the following covariates: tidal range, sediment type, 90-day mean temperature (prior to sampling), 90 day turbidity (prior to sampling), and mean turbidity during the flood period scaled by time since the flood squared (Table 4). Mean turbidity during the flood period was scaled so that its effect decays with time since the flood. Intuitively, this means that as the time since the flood increases, the impact of turbidity during the flood event decreases. We also considered scaling by time since flood (not squared), which produced similar results.

Table 4: Covariates in the final Moreton Bay seagrass cover model, their type (numeric or categorical), and if categorical, the number of unique levels.

Variable	Type	No. Levels
90 Day Temperature	Numeric	NA
90 Day Turbidity	Numeric	NA
Mean Flood Turbidity/Years Since Flood^2	Numeric	NA
Sediment Type	Categorical	6
Tidal Range	Categorical	3

Table 5: Fixed effects table of covariates in the Moreton Bay water quality and geographic model.

Term	Estimate	z	p.value
Intercept (Mud/Deep Subtidal)	-13.683	-2.312	0.021
Muddy Sand	4.825	5.674	0.000
Other	11.296	1.188	0.235
Rock	-8.661	-5.328	0.000
Sand	-0.095	-0.098	0.922
Sandy Mud	3.492	4.032	0.000
Intertidal	22.014	24.671	0.000
Shallow Subtidal	11.633	16.414	0.000
90 Day Temperature	0.876	4.119	0.000
90 Day Turbidity	-7.181	-2.802	0.005
Mean Flood Turbidity/Years Since Flood^2	-2.056	-3.459	0.001

Table 6: ANOVA table of covariates for the categorical variables in the Moreton Bay water quality and geographic model.

Effects	df	Chi-sq	p.value
Tidal Range	2	608.900	0.000
Sediment Type	5	113.720	0.000

Table 5 shows the fixed effects table from the fitted model. The reference group is the muddy and deep subtidal group, and the remaining sediment type and tidal range terms represent deviations from these baselines. For the numeric variables, all had large test statistics and small p-values (p-value < 0.01). While we can assess the overall significance of the numeric variables Table 5, we must perform an analysis of variance (ANOVA) to assess the overall significance of variables with many levels. Table 6 shows these ANOVA results for the sediment type and tidal range variables, which had large test statistics and small p-values (p-value < 0.01). Together, these results indicate all variables had a significant relationship with total seagrass cover. Here are some relevant takeaways from the model fit:

- Tidal Range: The estimated (marginal) average total cover is largest in the intertidal (26.89%), followed by shallow subtidal (16.51%) and deep subtidal (4.88%).
- Sediment Type: The estimated (marginal) average total cover is largest in the other category (25.6%), followed by muddy sand (19.1%), sandy mud (17.8%), mud (14.3%), sand (14.2%), and rock (5.62%). The other category results should be interpreted with caution, as there are only four observations.
- 90 Day Temperature: A one-degree (C) increase in 90-day water temperature prior to sampling is associated with a decrease in average seagrass cover by 0.88%.
- Mean Flood Turbidity/(Years Since Flood)^2: A one-unit increase in mean turbidity during the flood is associated with a decrease in average seagrass cover of 2.05%/(Years Since Flood)^2 (Figure 4).
- 90 Day Turbidity: A one-unit increase in 90-day turbidity (prior to sampling) is associated with a decrease in average seagrass cover by 7.18% (Figure 4).

We examined model diagnostics to better understand the overall model fit. First, we partitioned the model into different components of variability (Table 7) and found that 11.2% is attributable to the covariates (a quantity sometimes referred to as the Pseudo R-squared), 31.6% to the spatial random error, 7.3% to the EHMP subzone, and the remaining variability is independent random error. Second, we examined the decay behaviour of spatial autocorrelation. The estimated exponential spatial range (ϕ) is 745.5 meters, which suggests two total seagrass cover observations are approximately (spatially) uncorrelated at a distance of $3\phi \approx 2$km, or two kilometers. Third, we analyzed the standardized model residuals, which have been decorrelated, to assess model assumptions and identify influential observations that had a sizable impact on model fit. Generally, the standardized residuals are centered around zero, though there is a slight right skew suggesting some observed total cover values are larger

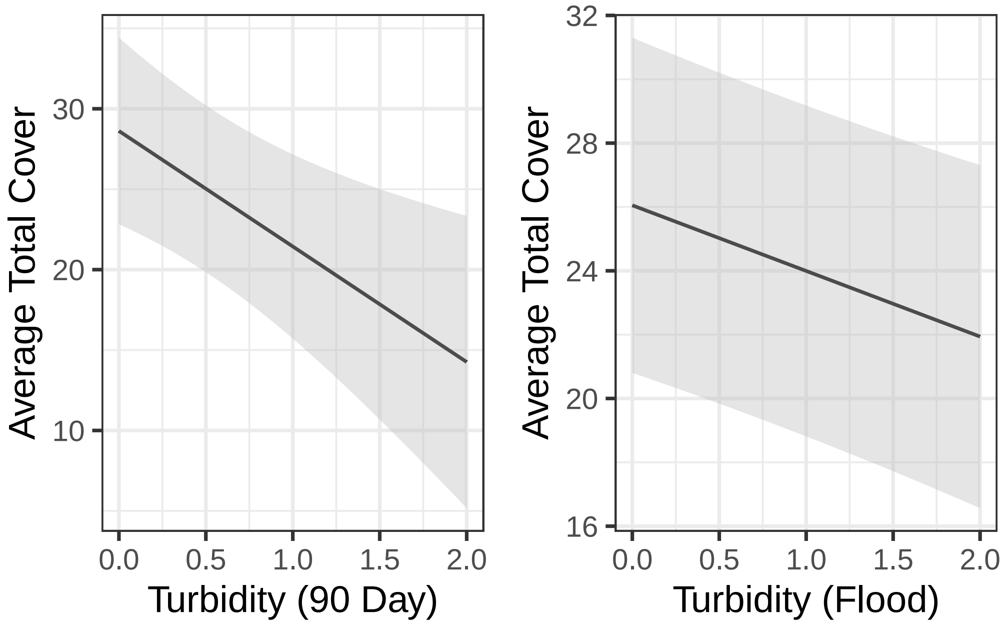

Figure 4: Average total seagrass percent cover as a function of 90-day mean turbidity (left) and flood turbidity (right). Estimates (and confidence intervals) were evaluated at one year post flood, at the means of all other numeric variables, and for the intertidal depth range and sand sediment types.

Table 7: Variance components in the Moreton Bay seagrass model.

Variance Component	Percent (of Total Variability)	Parameter Value
Covariates	11.2%	
Spatial Effect	31.6%	182.1
Independent Error	49.9%	287.5
EHMP Subzone	7.3%	745.5

than expected (according to the model). Observations are referred to as "influential" if they highly influence model fit. Influence is often assessed using a Cook's distance threshold value of one [Cook and Weisberg, 1982]. No observations exceed this threshold.

2.1.4 Block Kriging

A tremendous benefit of using the spatial linear model is that block Kriging [Ver Hoef, 2002, Cressie, 2006] can be used to predict the mean total seagrass cover bay-wide for any combination of covariates and geographic regions, even if these variables were not directly used in the model (see Appendix 1 for additional details). Block Kriging was undertaken, by financial year, for EHMP subzone, Marine Park zone, tidal range and for the whole of Moreton Bay.

Figure 5 and Figure 6 show there are several EHMP subzones and Marine Park zones that have recovered to pre-flood levels (e.g., Deception Bay, Marine National Park Zone), while others have not (e.g., Southern Bay, General Use Zone). When we examine recovery by tidal range (Figure 7), we see that seagrass cover is recovering across all tidal ranges. When we examine recovery for the whole of Moreton Bay (Figure 8), the model suggests that seagrass cover has improved since the 2022 flood.

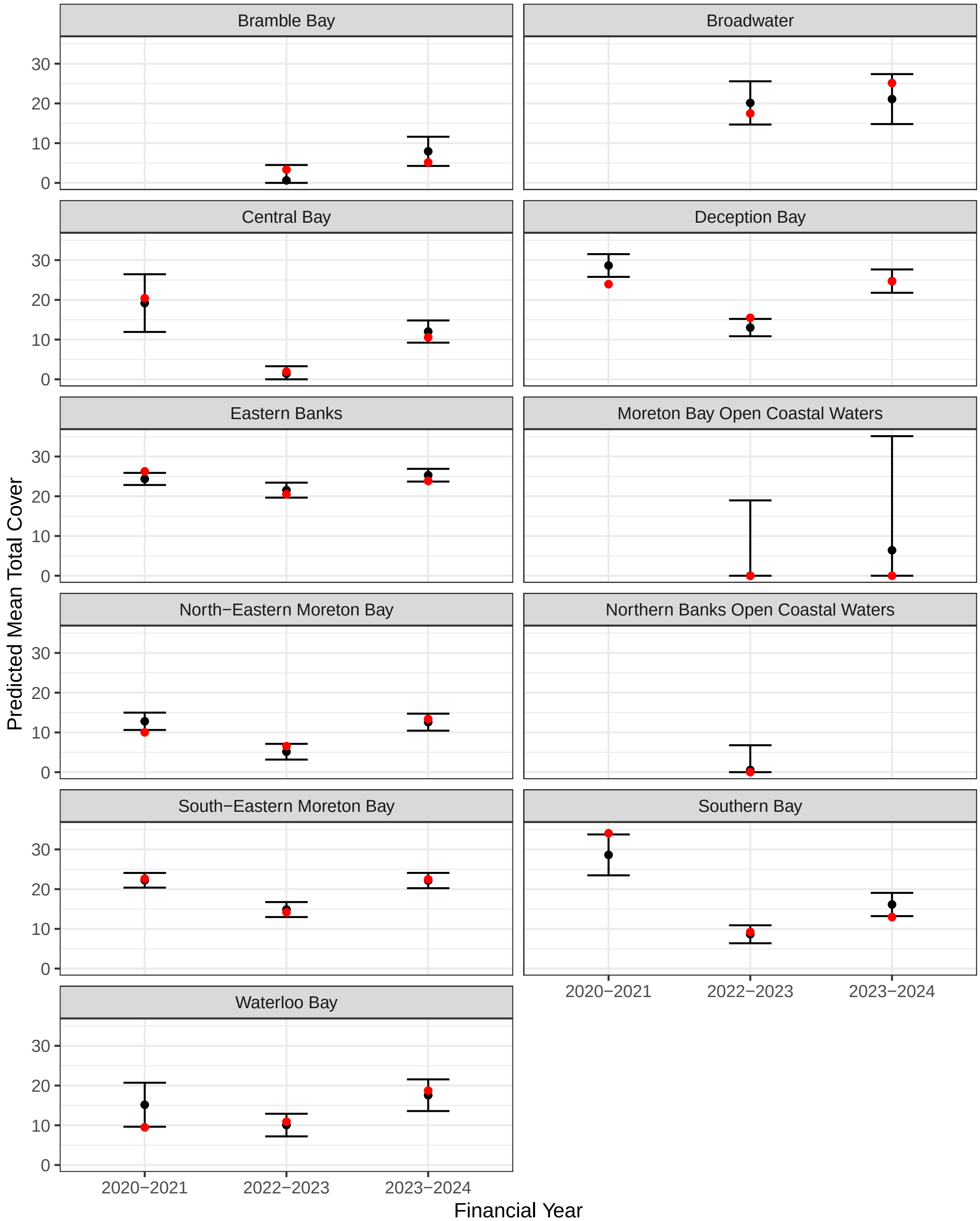

Figure 5: Average total cover predictions by EHMP subzone. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

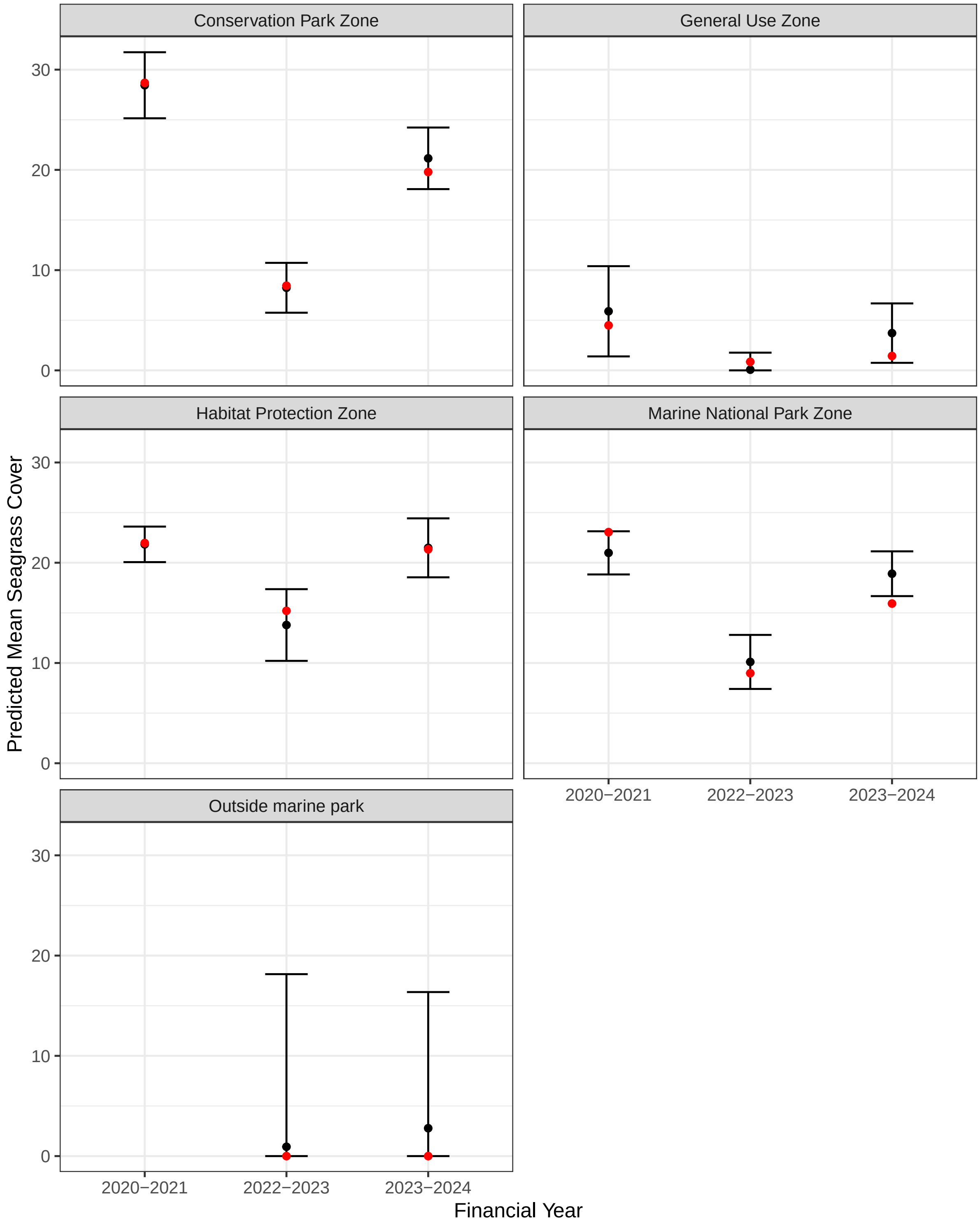

Figure 6: Average total seagrass cover predictions by conservation zone. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

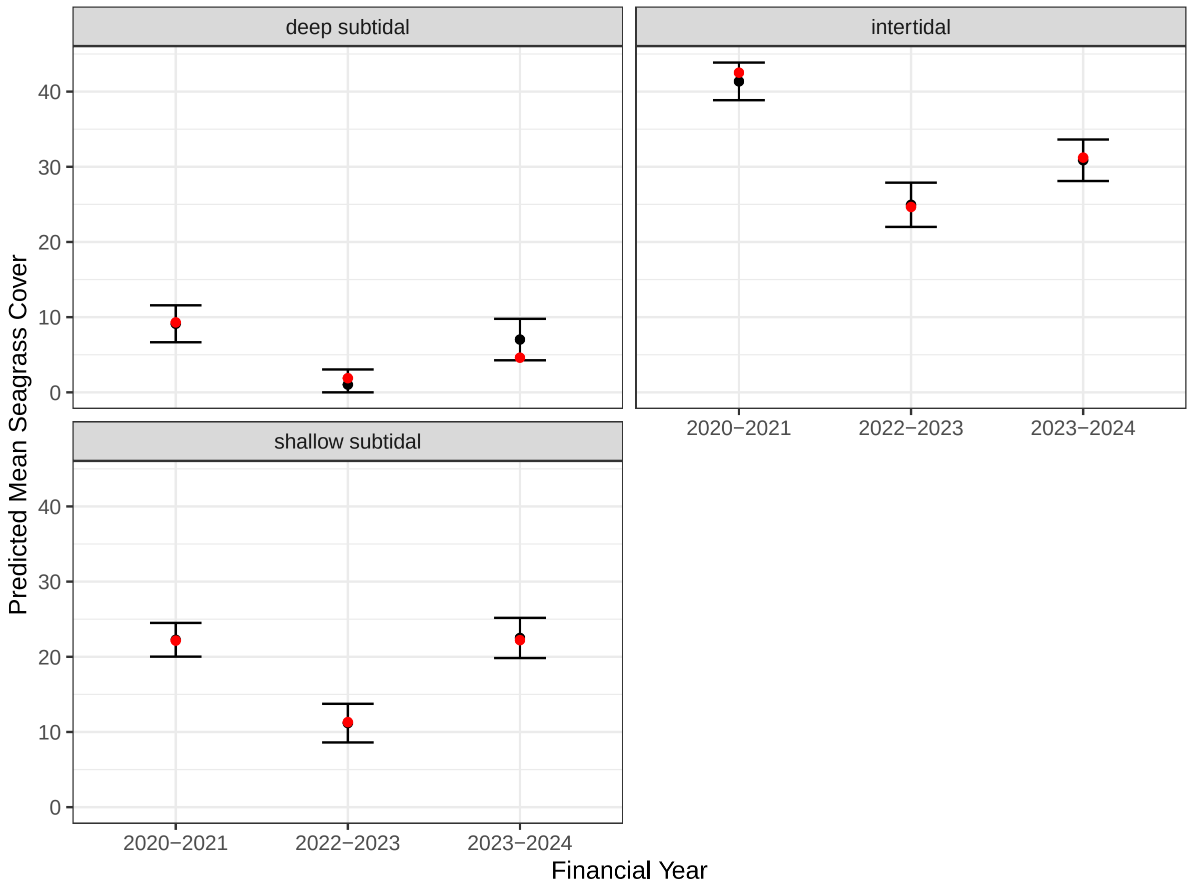

Figure 7: Average total seagrass cover predictions by tidal range. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

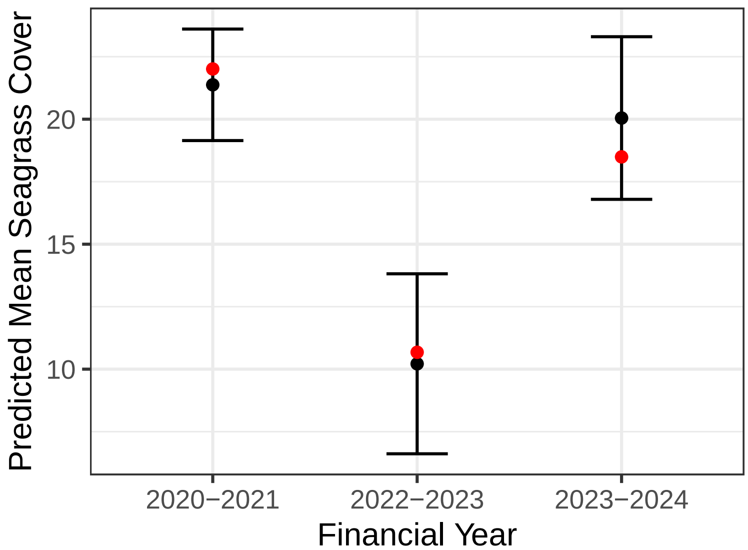

Figure 8: Average total seagrass cover predictions in Moreton Bay. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

3 Resilience Model

The seagrass cover model quantifies bay-wide effects of influential drivers on seagrass cover and enables assessment of recovery trajectories at management-relevant scales. However, it does not quantify site-specific resilience. Gilby et al. [2023] propose identifying hotspots (i.e., brightspots, sites performing better than expected) and coldspots (sites performing worse than expected) based on model residuals exceeding confidence interval bounds. They argue that hotspots and coldspots offer significant utility for understanding drivers of site-specific resilience and informing management decisions. Gilby et al. [2023] primarily focussed on specific hotspots, linking environmental variables to those sites. We build on that work by using the seagrass model residuals as a proxy for site-specific resilience, and then characterizing the most important drivers of that resilience.

3.1 Methods

3.1.1 Data

We used the raw seagrass model residuals (Figure 9) as a the response variable in a second spatial linear model used to study site-specific resilience in Moreton Bay. Model covariates included variables describing seagrass richness and dominant species, and bathymetry, as well as distance to the nearest coral, seagrass, mangrove, or gastropod habitat, and the total area in msq of coral, seagrass, mangrove, or gastropod habitat within 500 m of the observation point (Table 8). EHMP subzone was again included as a random effect. Because we were interested in hypothesis testing rather than parsimony, we included all potential covariates as fixed effects in the model. Potential covariates have the following structure:

- Coral Habitat Total Area is numeric from 0 msq to 616,180 meters squared (msq).
- Seagrass Habitat Total Areais numeric from 0 msq to 773,161 msq.
- Mangrove Habitat Total Area is numeric from 0 msq to 561,317 msq.
- Gastropod Habitat Total Area is numeric from 0 msq to 114,795 msq.
- Distance to Coral Habitat is numeric from 0 m to 35,387 m.
- Distance to Seagrass Habitat is numeric from 0 m to 11,382 m.
- Distance to Mangrove Habitat is numeric from 0 m to 27,617 m.
- Distance to Gastropod Habitat is numeric from 0 m to 32,034 m.
- Habitat Count is numeric (integer) from 0 to 4 unique habitats (coral, seagrass, mangrove, gastropod).
- Species Richness is numeric (integer) from 0 to 4 unique species (possible species: “Absent”, “CS”, “HD”, “HO”, “HS”, “SI”, and “ZMHU”).
- Dominant Species represents the dominant seagrass species and has seven levels: “Absent”, “CS”, “HD”, “HO”, “HS”, “SI”, and “ZMHU”.
- Bathymetry Mean Depth is the GBR mean depth within 500 m of the seagrass observation point, excluding terrestrial areas, and is numeric from -35.95 m to 0.79 m.

- Bathymetry Mean Curvature is the GBR mean curvature within 500 m of the seagrass observation point, excluding terrestrial areas, and is numeric from -0.002 degrees to 0.0001 degrees.
- Bathymetry Mean Slope is the GBR mean slope within 500 m of the seagrass observation point, excluding terrestrial areas, and is numeric from 0.009 degrees to 4.27 degrees.
- EHMP subzone is a categorical variable with eleven levels: Bramble Bay, Broadwater, Central Bay, Deception Bay, Eastern Banks, Moreton Bay Open Coastal Waters, North-Eastern Moreton Bay, Northern Banks Open Coastal Waters, South-Eastern Moreton Bay, Southern Bay, and Waterloo Bay (Figure 2).

We expected there to be nonlinear relationships between the response and covariates and we accounted for them using B-spline basis functions [de Boor, 1978, Eilers and Marx, 1996]. The degree of these functions was informed by partial dependence plots derived from a random forest model fitted to the seagrass model residuals (Table 8).

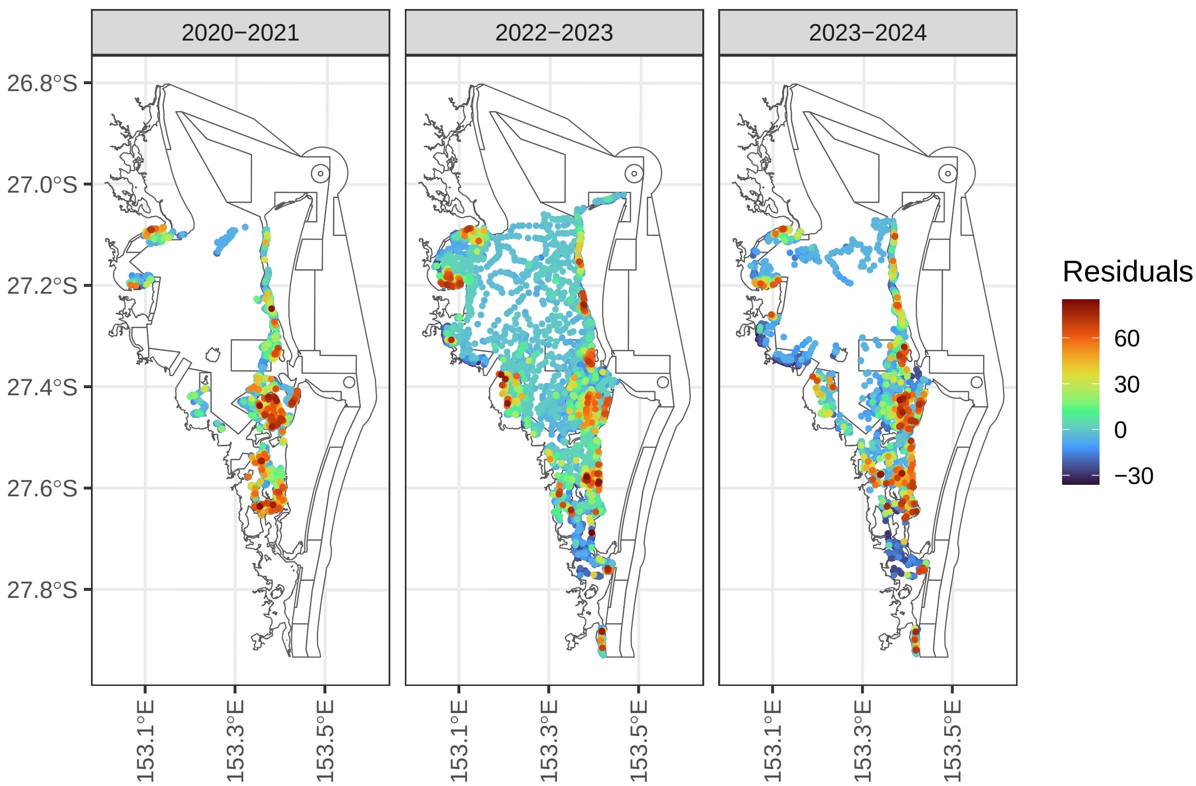

Figure 9: Seagrass model residuals for resilience analysis in Moreton Bay, by financial year.

Table 8: Covariates included in the resilience model, including whether they were a fixed or random effect, and the nonlinearity degree of the B-splines basis function.

Variable	Effect	Degree
Coral Habitat Total Area	Fixed	2
Seagrass Habitat Total Area	Fixed	2
Mangrove Habitat Total Area	Fixed	2
Gastropod Habitat Total Area	Fixed	2
Distance to Coral Habitat	Fixed	3
Distance to Seagrass Habitat	Fixed	2
Distance to Mangrove Habitat	Fixed	2
Distance to Gastropod Habitat	Fixed	2
Species Richness	Fixed	1
Habitat Count	Fixed	2
Dominant Species	Fixed	NA
Bathymetry Mean Depth	Fixed	2
Bathymetry Mean Curvature	Fixed	2
Bathymetry Mean Slope	Fixed	2
EHMP Subzone	Random	NA

3.1.2 Statistical Modeling

Table 9 shows the fixed effects table from the fitted model. The reference group is the absent dominant species, and the remaining dominant species terms represent deviations from this baseline. Species richness, the only numeric variable with a single level, had a large test statistic and small p-value (p-value < 0.01). The other numeric variables were modeled using a B-spline basis of at least degree two to capture nonlinearities. While we can assess the overall significance of species richness in Table 9, we must perform an analysis of variance (ANOVA) to assess the overall significance the remaining variables (which have at least two terms). Table 10 shows these ANOVA results; here are some relevant takeaways from both tables:

- Dominant Species: The estimated (marginal) average resilience score is largest for CS (38.4), followed by ZMHU (18.8), HS (10.8), SI (10.7), HO (3.7), and HD (-3.1).
- Species Richness: An increase of species richness by one species was associated with an average increase in resilience by approximately two units (Figure 10).
- Bathymetry Mean Depth: Deeper sites tended to be more resilient than shallower sites, on average (Figure 10).

We found moderate evidence (p-value < 0.1) that bathymetry mean curvature, gastropod habitat total area, and bathymetry mean slope are associated with resilience:

Table 9: Fixed effects table of covariates in the Moreton Bay resilience model.

Term	Estimate	z	p.value
Intercept (Absent)	-30.413	-2.588	0.010
CS	47.680	32.519	0.000
HD	6.182	1.071	0.284
HO	12.965	14.788	0.000
HS	20.025	21.705	0.000
SI	19.904	4.908	0.000
ZMHU	28.048	32.837	0.000
Species Richness	1.911	4.710	0.000
Coral Area Deg1	-8.420	-1.620	0.105
Coral Area Deg2	1.939	0.272	0.786
Seagrass Area Deg 1	-2.812	-1.367	0.171
Seagrass Area Deg1	-0.785	-0.500	0.617
Mangrove Area Deg1	-3.105	-0.729	0.466
Mangrove Area Deg2	1.795	0.298	0.766
Gastropod Area Deg1	10.378	1.099	0.272
Gastropod Area Deg2	-22.961	-2.238	0.025
Habitat Count Deg1	-1.428	-0.488	0.625
Habitat Count Deg2	1.816	0.649	0.516
Coral Distance Deg1	-5.043	-0.615	0.539
Coral Distance Deg2	0.161	0.018	0.985
Coral Distance Deg3	0.999	0.131	0.895
Seagrass Distance Deg1	7.656	1.232	0.218
Seagrass Distance Deg2	-11.543	-1.110	0.267
Mangrove Distance Deg1	-9.280	-1.298	0.194
Mangrove Distance Deg2	7.514	0.918	0.359
Gastropod Distance Deg1	0.772	0.119	0.905
Gastropod Distance Deg2	-3.009	-0.749	0.454
Bathymetry Mean Depth Deg1	21.332	2.129	0.033
Bathymetry Mean Depth Deg2	-2.460	-0.374	0.708
Bathymetry Mean Slope Deg1	3.698	1.005	0.315
Bathymetry Mean Slope Deg2	-9.865	-1.930	0.054
Bathymetry Mean Curvature Deg1	16.126	1.398	0.162
Bathymetry Mean Curvature Deg2	23.319	2.568	0.010

Table 10: ANOVA table of covariates in the Moreton Bay resilience model.

Effects	df	Chi-sq	p.value
Dominant Species	6	1637.726	0.000
Bathymetry Mean Depth	2	21.876	0.000
Bathymetry Mean Curvature	2	8.999	0.011
Gastropod Habitat Total Area	2	5.155	0.076
Bathymetry Mean Slope	2	4.768	0.092
Coral Habitat Total Area	2	2.845	0.241
Seagrass Habitat Total Area	2	1.971	0.373
Distance to Mangrove Habitat	2	1.871	0.392
Distance to Seagrass Habitat	2	1.836	0.399
Habitat Count	2	0.722	0.697
Distance to Gastropod Habitat	2	0.613	0.736
Distance to Mangrove Habitat	2	0.536	0.765
Distance to Coral Habitat	3	0.718	0.869

- Bathymetry Mean Curvature: Sites with more curvature tended to be less resilient than sites with less curvature.
- Gastropod Habitat Total Area: Sites with more gastropod habitat area tended to be more resilient than sites with less gastropod habitat area.
- Bathymetry Mean Slope: Sites with less extreme slopes (between zero and one) tended to have increasing resilience with slope, while sites with more extreme slopes (greater than one) tended to have decreasing resilience with slope.

We found little evidence (p-value > 0.1) the remaining variables were associated with resilience, after controlling for other model covariates.

As mentioned previously, we also fit a random forest model to the seagrass model residuals. Largely, the results were similar to the spatial linear model. Notably, for the random forest:

- Dominant species was most important, with a similar ranking of resilient species as for the spatial linear model approach (e.g., CS, ZMHU most resilient and Absent least resilient).
- Species richness was second-most important, with a general increasing trend in resilience with richness.
- Distance to gastropod habitat was third-most important, with a decreasing trend at relatively short distances ($< 15{,}000$ m) and an increasing trend at large distances ($> 15{,}000$ m).
- Bathymetry mean depth was the fourth-most important variable, with a general decreasing trend with decreasing depth (Figure 11), which is similar to the spatial linear model results.

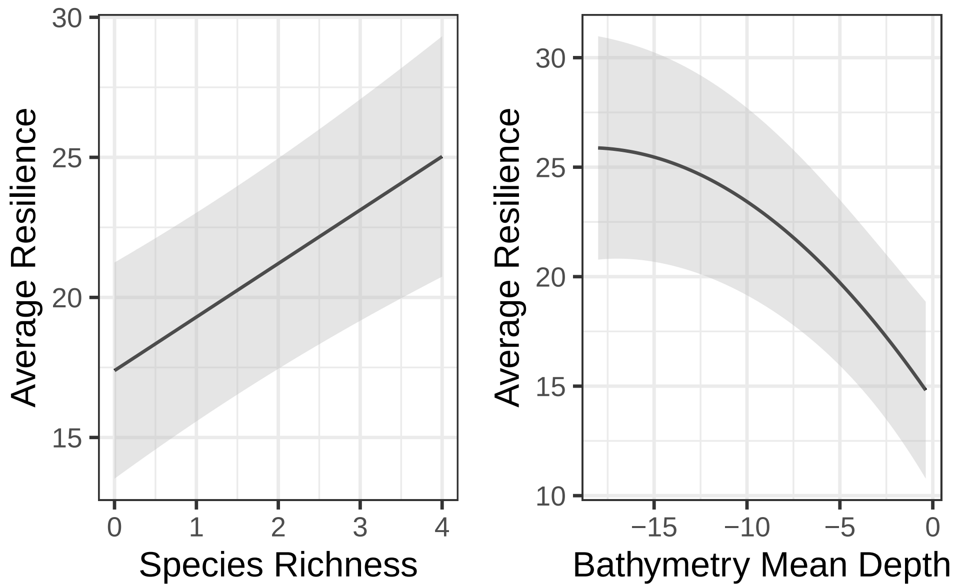

Figure 10: Average species richness (left) and bathymetry mean depth (right) effects on resilience. Predictions (and confidence intervals) were evaluated at the means of all other numeric variables and for the ZMHU species in the Southern Bay (similar trends occur for both variables among the remaining species and subzones).

- Seagrass area, distance to mangrove habitat, and distance to coral habitat had similar importance as the remaining bathymetry variables.
- The remaining variables (distance to seagrass habitat, mangrove area, EHMP subzone, habitat count, coral area, and gastropod area) were least important.

The consistency between the two resilience models and their variable importance rankings reinforce the spatial linear model's identification of dominant species, species richness, and bathymetry mean depth as key resilience drivers.

4 Limitations

While the spatial linear model was useful for characterizing broad patterns in seagrass cover related to tidal range, sediment type, temperature, and turbidity and site-specific patterns capturing resilience, all while accounting for spatial dependence, there are some limitations. First, the eReefs water-quality covariates are based on modeled data that are not directly observed in the field, potentially propagating modeling error. Second, there are many correlated water-quality variables, making it difficult for any model to isolate the effects of each component of water quality pre- and post-flood. Third, the lack of a covariate describing mean salinity in the 90 days prior to sampling makes it challenging to directly study the effect of

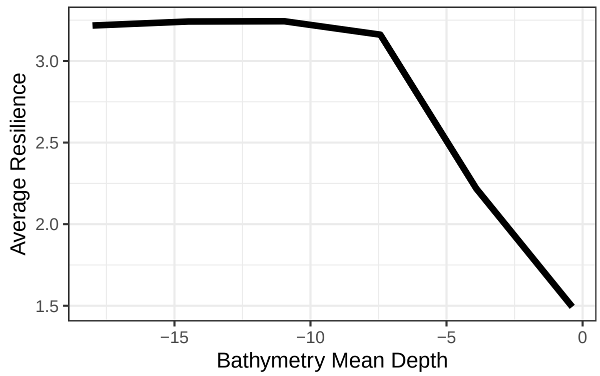

Figure 11: Random forest partial dependence for resilience as a function of bathymetry mean depth.

mean salinity during the flood period. Fourth, seagrass cover data was not collected in the first four months following the flood. It is therefore unclear whether the seagrass data collected in the 2022-2023 financial year fully describe flood impacts. Fifth, covariates representing local-scale drivers of seagrass cover (e.g., light attenuation, biological interactions, etc.) were not available, which might explain some additional variability in the response through space and time. Sixth, the location of sample sites was not generated randomly and could be subject to conscious or unconscious selection bias.

5 Revisiting the Research Questions

Research Question 1: Can we identify water quality covariates that act as triggers for seagrass cover management responses? Are these triggers different based on whether they result from a specific event (e.g., flood) or are ambient?

We identified ambient 90-day turbidity, turbidity during the flood event, and ambient 90-day water temperature as significant water quality covariates in the seagrass cover model. Our results suggest that ambient and flood-related turbidity act through distinct mechanisms and at different magnitudes; a one-unit increase in 90-day mean turbidity was associated with a 7.18 percentage point decrease in total cover, while the effect of flood turbidity decayed with time since the flood. Across tidal ranges, intertidal sites supported the highest average total cover (26.89%), followed by shallow subtidal (16.51%) and deep subtidal (4.88%), suggesting that management thresholds may need to be calibrated differently depending on depth zone. Similarly, sediment type affected baseline cover levels, as sites with muddy sand and sandy mud substrates supported higher average seagrass cover than sand or rock sites. Together, these results suggest that management triggers should differ depending on whether the driver is chronic ambient turbidity or an acute flood event, and should further account for the tidal and sediment context of the sites in question.

Research Question 2: Did seagrass cover improve after the flood event(s)? If so, how and where?

Total seagrass cover improved throughout Moreton Bay between 2022–2023 and 2023–2024, suggesting the system has been recovering since the flood. However, recovery was spatially heterogeneous; Deception Bay and the Marine National Park zone returned to pre-flood cover levels, while Southern Bay and the General Use Zone lagged considerably behind. Recovery was also relatively slow compared to the magnitude of the impact in the intertidal tidal range compared to shallow and deep subtidal ranges, suggesting that intertidal sites may warrant prioritized management attention immediately following floods.

Research Question 3: Can we identify site-specific resilience indicators, which describe where recovery is likely to be enhanced or impeded? What factors contribute to seagrass resilience?

We identified several site-specific indicators of seagrass resilience. Dominant species, species richness, and bathymetry mean depth were the strongest predictors of seagrass cover resilience, with sites dominated by CS or ZMHU species, higher species richness, and greater water depth tending to outperform model expectations and hence, infer greater resilience. Gastropod habitat area, bathymetric curvature, and slope provided moderate additional explanatory power. These findings may suggest that deeper, species-rich meadows dominated by CS and ZMHU exhibit a high level of resilience, while sites with low species richness and/or dominated by other species may require more active intervention to support recovery.

6 Summary

We applied a rigorous, spatially explicit statistical framework to characterize the drivers, recovery, and resilience of seagrass cover in Moreton Bay across a three-year monitoring period that included the 2022 Moreton Bay flood event. The analytical approach taken here is well-suited to the structure and scale of the data, and the findings have direct implications for adaptive management of the bay's seagrass ecosystem.

The use of spatial linear models as the primary analytical framework is well-justified from both statistical and ecological perspectives. Total percent seagrass cover observations collected across the bay are inherently spatially structured, as nearby sites share similar environmental conditions, water quality and disturbance histories, as well as ecological communities. Models that ignore this dependence may well provide misleading results and inference. By explicitly modeling spatial dependence via a spatial covariance function, the analyses presented here produce more reliable estimates of covariate effects and more honest characterizations of uncertainty than a conventional non-spatial model provides. These models are typically also more interpretable and characterize uncertainty more effectively than machine learning models. The inclusion of EHMP subzone as a random effect further accounts for structured variation among management-relevant regions of the bay, ensuring that covariate estimates reflect within-subzone patterns rather than being confounded by between-subzone differences in baseline cover. The block Kriging helps understand bay-wide trends and can immediately characterize bay-wide predictions on the original total seagrass cover scale (0-100), rather than a transformed scale required for generalized linear models; providing a flexible framework for scenario modeling under future flood conditions and climate scenarios.

The use of a two-stage modeling framework – first modeling total seagrass cover as a function of water quality and geographic drivers, then modeling the seagrass model residuals as a function of site-level resilience covariates – is a structured approach to separating bay-wide environmental patterns from site-specific ecological resilience capacity. This builds directly on the framework proposed by Gilby et al. [2023] to help understand drivers of average resilience and provide useful insight needed to inform broader conservation strategies. The robustness of the findings is further supported by the consistency of results across multiple modeling approaches (e.g., spatial beta regression model, random forest, etc.), which provides evidence that the relationships identified are genuine rather than resulting solely from specific modeling frameworks.

Overall, the results described in this report provide a quantitative, spatially structured foundation for adaptive decision making. Together, the identification of important water quality and geographic variables, bay-wide predictions via block Kriging, and the characterization of site-level resilience indicators provide data-driven evidence that can inform temporally targeted, context-specific interventions throughout Moreton Bay.

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7 Acknowledgements

The seagrass cover dataset was provided by Science Under Sail Australia (SUSA; https://www.scienceundersail.com.au/). The eReefs datasets (model simulations, satellite or in-situ observations) and software were produced as part of the eReefs project (ereefs.org.au), which is a collaboration between Australia's national science agency CSIRO, the Australian Institute of Marine Science (AIMS) and the Queensland Government, with observations obtained through the Integrated Marine Observing System (IMOS) and the Great Barrier Reef Marine Park Authority (GBRMPA). eReefs is funded by the Australian Government's Reef Trust.

Appendix 1. Spatial Linear Models and Random Forest Models

The spatially explicit statistical models (i.e., spatial models) we used to answer these research questions are related to commonly used linear and generalized linear models but incorporate *spatial dependence*. Spatial dependence measures how similar two sites are based on their distance. Formally, we can build spatial dependence directly into spatial models by leveraging Tobler's First Law of Geography [Tobler, 1970], which states succinctly that "nearby sites tend to be more similar than distant sites". Not only is this intuitive from a "first principles" perspective, but spatial models also provide many advantages for modeling spatial data compared to models that assume independence among observations (i.e., nonspatial models). First, nonspatial models tend to underestimate uncertainty, yielding covariate standard errors and p-values for covariates that are too small and corresponding confidence intervals that are too narrow. Second, spatial models improve prediction at unsampled locations by borrowing strength from nearby observations, yielding more precise predictions and corresponding standard errors that vary by location. Third, spatial models decompose model error into separate spatial and nonspatial components, informing residual and resilience structure in ways that nonspatial models cannot. Fourth, spatial models can provide ecologically meaningful insight into the spatial dependence structure of the process, informing relationships among measured and unmeasured variables as well as future sampling plans. Zimmerman and Ver Hoef [2024] provide a thorough review of spatial models for ecological and environmental data, providing more detail and justification for the claims made above.

Spatial models also offer many benefits over commonly used machine learning algorithms like random forests [Breiman, 2001, Cutler et al., 2007, James et al., 2017, Kuhn and Silge, 2022]. First, traditional machine learning algorithms do not incorporate spatial dependence and, by consequence, often perform worse than spatial models at characterizing important covariates and predicting at unobserved locations [Fox et al., 2020]. Second, machine learning algorithms struggle to quantify uncertainty in a principled manner, something spatial models excel at. Third, covariate inference is more straightforward using spatial models, as they naturally provide p-values that enable structured testing of hypotheses. Fourth, spatial models provide a rich set of diagnostics like leverage and Cook's distances [Montgomery et al., 2021], quantities not generally defined for machine learning algorithms. This is not to say that machine learning algorithms cannot be very useful in ecological data analyses, but rather that spatial models are sometimes a more effective tool for modeling spatial data.

Though there are many benefits of spatial models over machine learning algorithms, the reverse is also true. It is much simpler to capture complex nonlinearities using "out-of-the-box" machine learning algorithms, bypassing the tuning often required for spatial models that involve nonlinear components like splines. This can be especially useful for ecological data, which often exhibit nonlinear relationships [Fox et al., 2017]. Machine learning algorithms also typically rely on few, if any, distributional assumptions and can be more robust against unusual observations like outliers [Hastie et al., 2009]. Machine learning algorithms also tend to be much more computational efficient than spatial models. Importantly, machine learning algorithms and spatial models can be complementary tools that help shed insight into complicated ecological processes (something we reiterate in the upcoming analyses). An exciting and developing area of statistical research involves incorporating spatial modeling strategies directly into machine learning algorithms [Saha et al., 2023, Heaton et al., 2025].

The first spatially explicit model we define is the spatial linear model. The spatial linear model is a *mixed-effects model* [Pinheiro and Bates, 2000, Bolker, 2015] written as

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + ... + \beta_p x_{p,i} + \tau_i + \epsilon_i, \tag{1}$$

where $i = 1, 2, ..., n$ indexes the n observations. The quantity y_i is the response variable (for observation i), the β parameters are slope parameters controlling the covariates, $x_{j,i}$ (for $j = 1, 2, ..., p$). The random error τ_i is a spatial random error, and the random error ϵ_i is the nonspatial (i.e., independent) random error. The spatial random error is spatially structured, capturing spatial residual effects like environmental gradients we do not have data to fully represent. The nonspatial random error is not spatially structured and captures nonspatial residual effects like measurement error and microscale variation. The difference between a traditional (i.e., nonspatial) linear model and a spatial linear model is the inclusion of τ_i (in the spatial linear model). The spatial random error implied by τ_i is governed by a *spatial covariance function*. The spatial covariance function allows the model to incorporate spatial dependence, yielding the improved model performance previously described. Figure 12 shows three different spatial covariance functions that describe different ways the spatial covariance decays with distance. Model parameters and standard errors are estimated from data using likelihood-based [Patterson and Thompson, 1971, Harville, 1977, Wolfinger et al., 1994] or semivariogram-based [Cressie, 1985, Curriero and Lele, 1999] estimation methods. Often, the spatial linear model in Equation 1 is written more compactly using *matrix notation*:

$$\mathbf{y} = \mathbf{X}\beta + \tau + \epsilon, \tag{2}$$

where $\mathbf{X}$ is a matrix that holds the j covariates for each of n observations, and $\mathbf{y}$, β, τ, and ϵ are vectors that contain each of the n or j quantities from Equation 1. The spatial linear model can be adjusted to account for nonspatial random effects just as one would add random effects to a nonspatial linear mixed model:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \tau + \epsilon, \tag{3}$$

where $\mathbf{Z}$ is a random effect design matrix (dimension $n \times k$) that indexes each of the k random effects in $\mathbf{u}$. For more information about specific forms of various spatial covariance functions, see Schabenberger and Gotway [2017] and Zimmerman and Ver Hoef [2024].

Like the traditional linear model, the spatial linear model can capture complex nonlinearities in the data through the use of interaction effects [Faraway, 2002, Lenth and Piaskowski, 2025], splines [Chambers and Hastie, 1992, Ruppert et al., 2003], additive models [Wood, 2017, Pedersen et al., 2019], and penalized regression models like the ridge and lasso [Tibshirani, 1996].

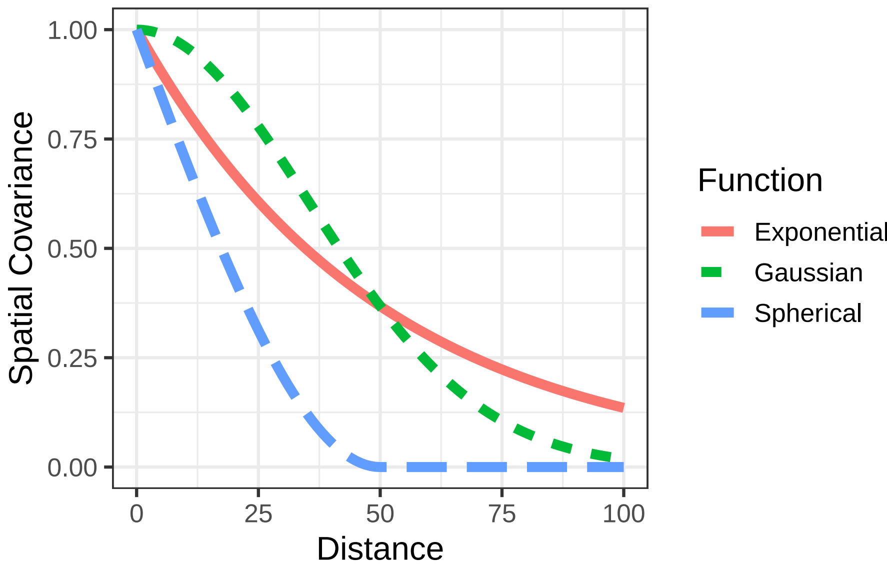

Figure 12: Three spatial covariance functions that represent different spatial dependence behavior.

Generally, this is accomplished by leveraging *basis functions* that break down nonlinear behavior into a set of additive linear components that may include penalty terms which guard against overfitting. Furthermore, there is no assumption that y_i itself is normally distributed; rather, distributional assumptions are placed on the random errors after incorporating the slope parameters. This clarification implies the spatial (and nonspatial) linear models can be quite effective at modeling a range of data that are not normally distributed. For highly nonnormal data, tools like generalized linear models, which we discuss later, can be helpful. Finally, under general conditions, the slope parameters of the spatial (and nonspatial) linear models are approximately normally distributed with a large sample size (see Casella and Berger [2024] for a general justification via the Central Limit Theorem and Zhang and Zimmerman [2005] for nuances regarding spatial data). This is beneficial because it justifies valid hypothesis testing on the slope coefficients (often of primary ecological interest), no matter the distribution of the data (given certain assumptions are satisfied).

One of the most useful applications of the spatial linear model is predicting the response variable at unobserved locations. A very useful spatial prediction method called Kriging has a rich theoretical justification [Cressie, 1990, Stein, 1999]. Kriging is the spatial analogue to best linear unbiased prediction (BLUP) for mixed models [Henderson, 1975]. Via Kriging, the spatial linear model produces BLUPs at each unobserved location of interest, alongside prediction intervals to characterize uncertainty. While useful, sometimes the goal is to create a BLUP for an entire region of interest. This technique, which builds upon the theoretical foundations of Kriging, is called block Kriging [Ver Hoef, 2002, Cressie, 2006]. The intuition is to make point predictions at a very dense grid in the study region and then average these

Table 11: Common generalized linear model families, link functions, and appropriate data types.

Family	Link Function	Link Name	Data Type
Binomial	$f(\mu) = \log(\mu/(1-\mu))$	Logit	Binary; Binary Count
Poisson	$f(\mu) = \log(\mu)$	Log	Count
Negative Binomial	$f(\mu) = \log(\mu)$	Log	Count
Beta	$f(\mu) = \log(\mu/(1-\mu))$	Logit	Proportion
Gamma	$f(\mu) = \log(\mu)$	Log	Skewed
Inverse Gaussian	$f(\mu) = \log(\mu)$	Log	Skewed

predictions, appropriately characterizing uncertainty. Together, Kriging and block Kriging provide a complete set of tools for spatial prediction, given the fit of a spatial linear model.

Spatial generalized linear models are an extension of spatial linear models to highly nonnormal data [Christensen and Waagepetersen, 2002, Zhang, 2002, Diggle and Ribeiro Jr, 2007, Bonat and Ribeiro Jr, 2016]. Ver Hoef et al. [2024] proposed a novel application of the Laplace approximation to enable restricted maximum likelihood estimation of spatial generalized linear models. This model formulation builds upon Equation 1 and is given by

$$g(\mu_i) = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + ... + \beta_p x_{p,i} + \tau_i + \epsilon_i, \tag{4}$$

where $g(\mu_i)$ is a "link function" that links the distribution of y_i to a linear function of its mean, μ_i. Common link functions include the log link for count and skewed data and the logit link for binomial and proportion data (Table 11). Using the same argument as for spatial linear models, nonspatial random effects can be added to spatial generalized linear models.

Spatial models are complicated, both theoretically and computationally. This has contributed to their relative underuse in the ecological community. However, recent advances in open-source software have made spatial models more accessible to ecologists who have strong subject matter expertise but may lack sufficient statistical background or computational skills to implement them from scratch. One such advancement is the `spmodel` **R** package [Dumelle et al., 2023], which emulates base-**R** functions like `lm()` and `glm()` to account for spatial dependence via `splm()` and `spglm()`, respectively. Familiar **R** functions like `summary()` and `predict()` can be used directly on models fit via `spmodel`. Importantly, `spmodel` also has tools for spatial random forest residual modeling [Fox et al., 2020], applications to large data of several thousand observations via spatial indexing [Ver Hoef et al., 2023], and cross validation [Stone, 1974, Roberts et al., 2017].